

Unit 8: The Derivative as a Function

1. The idea, in plain words

In Unit 7, you computed $f'(a)$ — the slope at one single, specific point. That’s a little like measuring the steepness of a hiking trail at just one particular mile marker. This unit asks: why stop at one point? Instead of plugging in a fixed number a , we keep x as a variable all the way through the computation. What comes out the other end isn’t just one number anymore — it’s a whole new function, $f'(x)$, that tells you the slope at every single point at once.

Picture it: if $f(x)$ is a hiking trail’s elevation, then $f'(x)$ is a second trail map that shows you the steepness at every mile marker, all in one shot. Plug in any x -value into $f'(x)$, and out pops the slope of the original trail at exactly that spot — no need to redo the whole limit computation from scratch every time.

Reading the slope-function off the shape of the graph: since $f'(x)$ is just “the slope of f at x ,” you can often tell a lot about f' just by looking at a picture of f :

- Where f is going uphill (increasing), $f'(x)$ is positive.
- Where f is going downhill (decreasing), $f'(x)$ is negative.
- Where f has a peak or a valley (a turning point with a flat tangent line), $f'(x) = 0$ right there.

An important one-way street: if a function is differentiable at a point, it’s guaranteed to also be continuous there — you can’t have a well-defined tangent slope at a point where the graph is literally broken. But the reverse isn’t true! A function can be perfectly continuous (no breaks) and still fail to be differentiable, if it has a sharp corner (like $|x|$ at $x = 0$, from Unit 7). So: differentiable $\Rightarrow$ continuous, but continuous does NOT $\Rightarrow$ differentiable. This gives you a handy shortcut: if you can show a function isn’t even continuous somewhere, you instantly know it can’t be differentiable there either — no need to compute any derivative limit at all.

2. Toolbox

The general definition (same formula as Unit 7, but now x stays a variable instead of being replaced by a fixed number):

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Notation — these all mean exactly the same thing:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx}$$

The one-way theorem:

$$\text{differentiable at } a \implies \text{continuous at } a$$

(but continuous at a does NOT guarantee differentiable at a)

Useful shortcut from the theorem (the contrapositive): if f is not continuous at a , then f is automatically not differentiable at a either — you can stop right there without any further computation.

Reading a graph: f increasing $\Leftrightarrow f' > 0$; f decreasing $\Leftrightarrow f' < 0$; f has a horizontal tangent (peak, valley, or flat spot) $\Leftrightarrow f' = 0$.

3. Common mistakes

- Doing the algebra as if x were a specific fixed number. Since x is now a variable, terms like x^2 , $2xh$, etc. must stay in the expansion — don't accidentally treat x like a known constant and drop it.
- Reading “high” as “steep.” A function's height on the graph (how big $f(x)$ is) has nothing to do with its slope (how big $f'(x)$ is). A function can be very high up but momentarily flat (small f'), or low down but very steep (large f').
- Assuming continuous automatically means differentiable. Remember, this only goes one direction. Always check separately if you suspect a corner or a break in the tangent behavior.
- Confusing $\frac{dy}{dx}$ with an actual fraction “dy divided by dx.” For now, treat it as a single symbol meaning “the derivative of y with respect to x ” — it behaves a bit like a fraction later on, but right now it's just notation.
- Forgetting that once you have the general formula $f'(x)$, you can plug in any point instantly — there's no need to redo the whole limit computation for every new point you want to evaluate.

4. Problem Set

Warm-up

1. Let $f(x) = x^2$. Use the definition to find the general formula for $f'(x)$.
2. Let $f(x) = 3x + 7$. Use the definition to find $f'(x)$.
3. Let $f(x) = x^2 - 4x$. Use the definition to find $f'(x)$.
4. Let $f(x) = 5 - 2x^2$. Use the definition to find $f'(x)$.
5. Let $f(x) = x^2 + 3x - 1$. Use the definition to find $f'(x)$, and also write your answer using $\frac{dy}{dx}$ notation.
6. Let $f(x) = 7$ (a constant function). Use the definition to find $f'(x)$, and explain in plain words why a constant function's derivative is always 0.

Standard

7. Let $f(x) = x^2 - 6x + 5$. Find $f'(x)$ using the definition. Then evaluate $f'(0)$, $f'(3)$, and $f'(6)$, and describe in words what's happening to the slope of the graph as x increases from 0 to 6.
8. Let $f(x) = -x^2 + 8x$. Find $f'(x)$ using the definition, then find the x -value where $f'(x) = 0$ (the location of the horizontal tangent).
9. Let $f(x) = x^3$. Use the definition to find $f'(x)$. (You'll need to expand $(x+h)^3$.)
10. You're told that for $f(x) = x^2$, the general derivative formula is $f'(x) = 2x$ (matching Problem 1). Without redoing the limit computation, quickly find $f'(5)$, $f'(-3)$, and the slope of the tangent line to f at $x = 5$.
11. Let $f(x) = 2x^2 + 3x$. Find $f'(x)$ using the definition, then find all x -values where $f'(x) = 7$.
12. A graph rises, then reaches a peak, then falls, then reaches a valley, then rises again. Describe, in words, where $f'(x)$ is positive, where it's negative, and where it equals 0.

Challenge

13. Let $f(x) = \frac{1}{x}$. Use the definition to find the general formula for $f'(x)$.

14. Let $f(x) = \sqrt{x}$. Use the definition to find the general formula for $f'(x)$. (You'll need the conjugate trick.)
15. Let $f(x) = \begin{cases} x+1, & x < 2 \\ x^2, & x \geq 2 \end{cases}$. Using the continuity-differentiability theorem as a shortcut (rather than computing any derivative limit), determine whether f could possibly be differentiable at $x = 2$. Explain your reasoning.
16. Let $f(x) = \begin{cases} 3, & x \leq 1 \\ 2x+1, & x > 1 \end{cases}$. First check whether f is continuous at $x = 1$. If it is, you'll need to go further and check differentiability directly — do so, and state your conclusion.
17. Let $f(x) = |x| + x^2$. Determine whether f is differentiable at $x = 0$ by computing the left-hand and right-hand difference-quotient limits.

Applied

18. A ball is thrown, and its height in meters after t seconds is $h(t) = -5t^2 + 20t + 2$. Find the general velocity function $v(t) = h'(t)$ using the definition. Then use your formula to find the velocity at $t = 0$, $t = 1$, and $t = 2$ seconds, and describe what's physically happening to the ball at each of those moments.
19. A company's revenue (in dollars) from selling x units is $R(x) = 100x - 0.5x^2$. Find the general marginal revenue function $R'(x)$ using the definition. Then evaluate $R'(50)$, $R'(100)$, and $R'(150)$, and describe the trend.
20. The number of people who've heard a rumor by day t is modeled by $N(t) = 200t - 4t^2$. Find the general rate function $N'(t)$ using the definition. Then evaluate $N'(10)$ and $N'(25)$, and interpret what each number means, including what's special about day 25.
21. An object's position in meters after t seconds is $s(t) = t^3 - 3t$. Find the general velocity function $s'(t)$ using the definition. Then evaluate $s'(0)$, $s'(1)$, and $s'(2)$, and describe the object's motion at each moment.